Engs 31 / CoSc 56 Final Project Report
RiSC-16 CPU

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Abstract

<Abstract text.>

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1 System Overview

1.1 Top-level Block Diagram

1.2 Description of Ports

Port	Dir.	Width	Description
clk	In	1	100 MHz system clock
<port>	In	<n>	<description>
<port>	Out	<n>	<description>

Table 1: Top-level ports.

1.3 Description of Components

Controller <Short description.>

ALU <Short description.>

Memory <Short description.>

Register File <Short description.>

Program Counter <Short description.>

Instruction Register <Short description.>

UART Receiver <Short description.>

Bootloader Controller and Registers <Short description.>

Top Level <Short description.>

1.4 Supporting Software

Within the Bootloader di

1.4.1 Purpose and Motivation

1.4.2 Usage and Workflow

1.4.3 Input and Output Formats

1.4.4 Verification of the Script

2 Technical Description

2.1 Controller

2.1.1 Description of Ports

Port	Dir.	Width	Description
<port>	<In/Out>	<n>	<description>

Table 2: <Component 1> ports.

2.1.2 Register Transfer Level (RTL) Diagram

2.1.3 Finite State Machine (FSM) Diagram

2.1.4 Description of Memory

2.2 ALU

2.2.1 Description of Ports

Port	Dir.	Width	Description
<port>	<In/Out>	<n>	<description>

Table 3: <Component 2> ports.

2.2.2 Register Transfer Level (RTL) Diagram

2.2.3 Finite State Machine (FSM) Diagram

2.2.4 Description of Memory

2.3 Memory

2.3.1 Description of Ports

Port	Dir.	Width	Description
<port>	<In/Out>	<n>	<description>

Table 4: <Component 3> ports.

2.3.2 Register Transfer Level (RTL) Diagram

2.3.3 Finite State Machine (FSM) Diagram

2.3.4 Description of Memory

2.4 Register File

2.4.1 Description of Ports

Port	Dir.	Width	Description
<port>	<In/Out>	<n>	<description>

Table 5: <Component 4> ports.

2.4.2 Register Transfer Level (RTL) Diagram

2.4.3 Finite State Machine (FSM) Diagram

2.4.4 Description of Memory

2.5 Program Counter

2.5.1 Description of Ports

Port	Dir.	Width	Description
<port>	<In/Out>	<n>	<description>

Table 6: <Component 5> ports.

2.5.2 Register Transfer Level (RTL) Diagram

2.5.3 Finite State Machine (FSM) Diagram

2.5.4 Description of Memory

2.6 Instruction Register

2.6.1 Description of Ports

Port	Dir.	Width	Description
<port>	<In/Out>	<n>	<description>

Table 7: <Component 6> ports.

2.6.2 Register Transfer Level (RTL) Diagram

2.6.3 Finite State Machine (FSM) Diagram

2.6.4 Description of Memory

2.7 UART Receiver

2.7.1 Description of Ports

Port	Dir.	Width	Description
<port>	<In/Out>	<n>	<description>

Table 8: <Component 7> ports.

2.7.2 Register Transfer Level (RTL) Diagram

2.7.3 Finite State Machine (FSM) Diagram

2.7.4 Description of Memory

2.8 Bootloader Controller and Registers

2.8.1 Description of Ports

Port	Dir.	Width	Description
<port>	<In/Out>	<n>	<description>

Table 9: <Component 8> ports.

2.8.2 Register Transfer Level (RTL) Diagram

2.8.3 Finite State Machine (FSM) Diagram

2.8.4 Description of Memory

3 Design Validation

3.1 <Component 1>

3.1.1 Behavioral Simulations

3.1.2 Hardware Validation

3.2 <Component 2>

3.2.1 Behavioral Simulations

3.2.2 Hardware Validation

3.3 <Component 3>

3.3.1 Behavioral Simulations

3.3.2 Hardware Validation

3.4 <Component 4>

3.4.1 Behavioral Simulations

3.4.2 Hardware Validation

3.5 <Component 5>

3.5.1 Behavioral Simulations

3.5.2 Hardware Validation

3.6 <Component 6>

3.6.1 Behavioral Simulations

3.6.2 Hardware Validation

3.7 <Component 7>

3.7.1 Behavioral Simulations

3.7.2 Hardware Validation

3.8 <Component 8>

3.8.1 Behavioral Simulations

3.8.2 Hardware Validation

3.9 <Subsystem 1>

3.9.1 <Subcomponent 1>

Behavioral Simulations

Hardware Validation

3.9.2 <Subcomponent 2>

Behavioral Simulations

Hardware Validation

3.10 Overall Design

3.10.1 Behavioral Simulations

3.10.2 Hardware Validation

4 Analysis of the Design

4.1 Resource Utilization

Resource	Used	Available	Utilization
Slice LUTs	<n>	20800	<%>
Slice Registers	<n>	41600	<%>
Block RAM Tile	<n>	50	<%>
DSPs	<n>	90	<%>
Bonded IOB	<n>	106	<%>
BUFGCTRL	<n>	32	<%>

Table 10: Post-implementation resource utilization (Artix-7 XC7A35T, Basys3).

4.2 Residual Warnings

Warning	Justification
<warning text>	<why it is safe to ignore>

Table 11: Residual warnings.

4.3 Division of Labor

4.4 Future Work

5 Acknowledgements

6 Conclusions

A VHDL Source Code

- `<file>.vhd` — <short description>
- `<file>.vhd` — <short description>

B VHDL Testbenches

- `<file>_tb.vhd` — <short description>
- `<file>_tb.vhd` — <short description>

C Support Script Source Code

- `<script>.py` — <short description>